

# Week 01. Part 1. Introduction

Population and Sample. Description and Inference.

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# Prerequisites overview

- Go to [Ilias MK68](#).

# The Population

- See: Ch. 1.3.3 Sampling from a population in ([Diez, Cetinkaya-Rundel, and Barr 2022](#))

# The Population

- “... is a natural, geographical, or political collection of people, animals, plants, or objects”
- “... is the **physical collection**, ... from which **we derive a set of values or the variable of interest**”
- specific to our **research question** and **the research design**
- defines the scope of our study.
- has quantifiable characteristics: true levels.

It is generally impossible to have data for the entire population

# Examples of a population by research questions

Discuss what are the (1) populations; (2) variables of interest; (3) true population value of the parameter based on the exemplary research questions below:

1. What is the average poverty headcount rate in the world?
2. What is the average wheat yield in 100 kg per ha?

# Examples of a population by research questions

At home, consider following research questions:

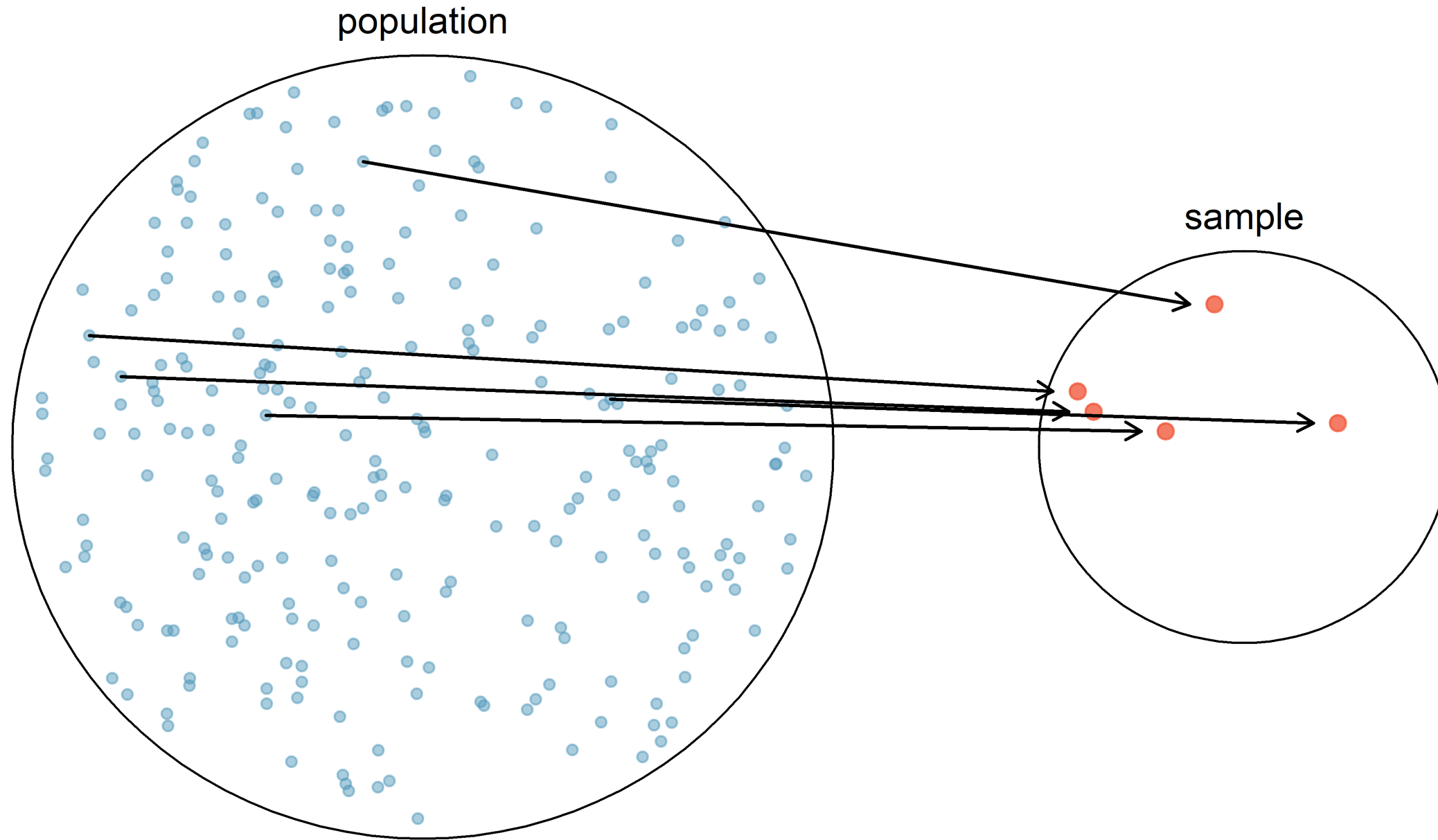
- How does access to education impact income inequality?
- What are the factors influencing malnutrition rates among children in a specific region?
- How does access to clean water and sanitation facilities impact child mortality rates?
- What are the barriers to accessing healthcare services among marginalized communities?

**How to actually answer such research questions?**

# The Sampling

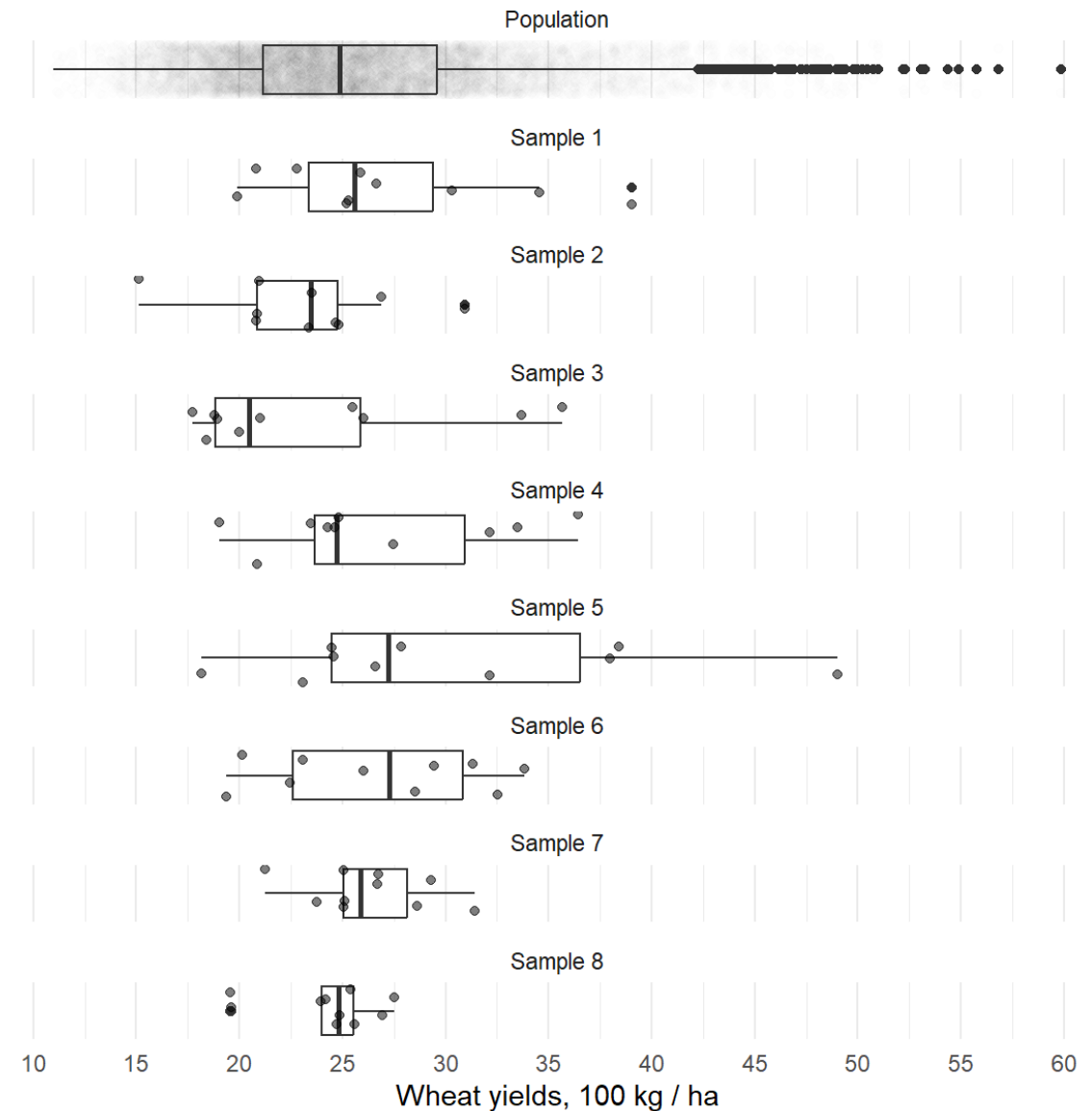
- See also book “Sampling” by ([Thompson 2012](#))
- The World Bank. Development Research in Practice (DIME)  
<https://dimewiki.worldbank.org/Sampling>

# Simple Random Sample (1)



# The Simple Random Sample (2)

- **Simple sample** is a random subset of observations from a population.
  - **Random** means
    - that every observation from the population has an **equal probability of being sampled**.
- Recall: What is the average wheat yield in 100 kg per ha?
- How different our samples are going to be?

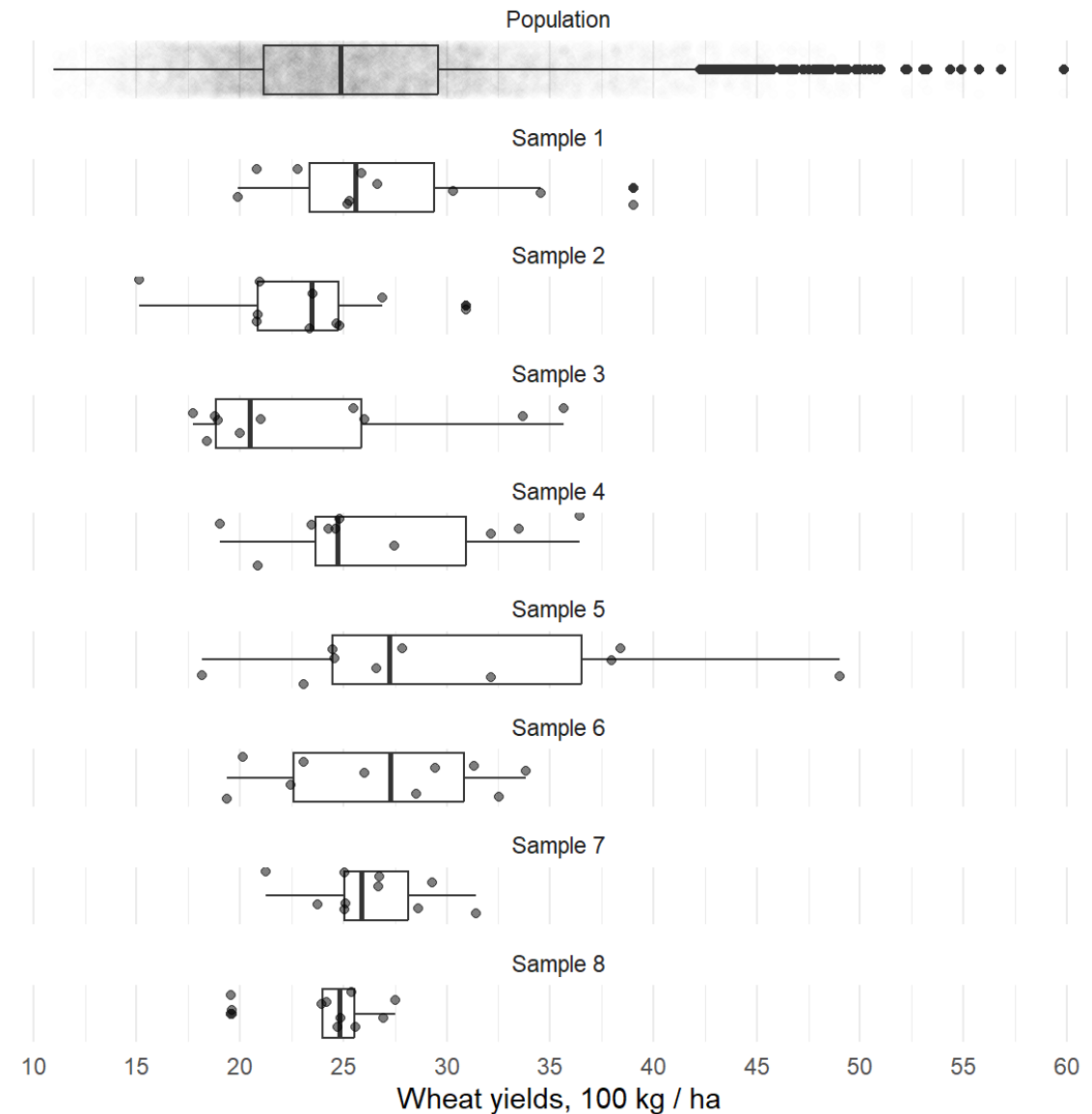


# The Simple Random Sample (3)

Each sample results with a different estimate of the yields:

| Type     | Yields |
|----------|--------|
| Sample 1 | 27.04  |
| Sample 2 | 23.19  |
| Sample 3 | 23.57  |
| Sample 4 | 26.66  |
| Sample 5 | 30.23  |
| Sample 6 | 26.68  |
| Sample 7 | 26.30  |
| Sample 8 | 24.23  |

- What is “**the correct**” average yields?
- Which sample estimates “**the correct**” average yield?
  - How to pick one?
  - Can we pick one?



# The power of a random sample

If every sample is unique, why random sample is so important?

A sample can approximate the population

- **when its size ( $N$ ) increases to infinity!**

This is the **Law of Large Numbers (LLN)**!

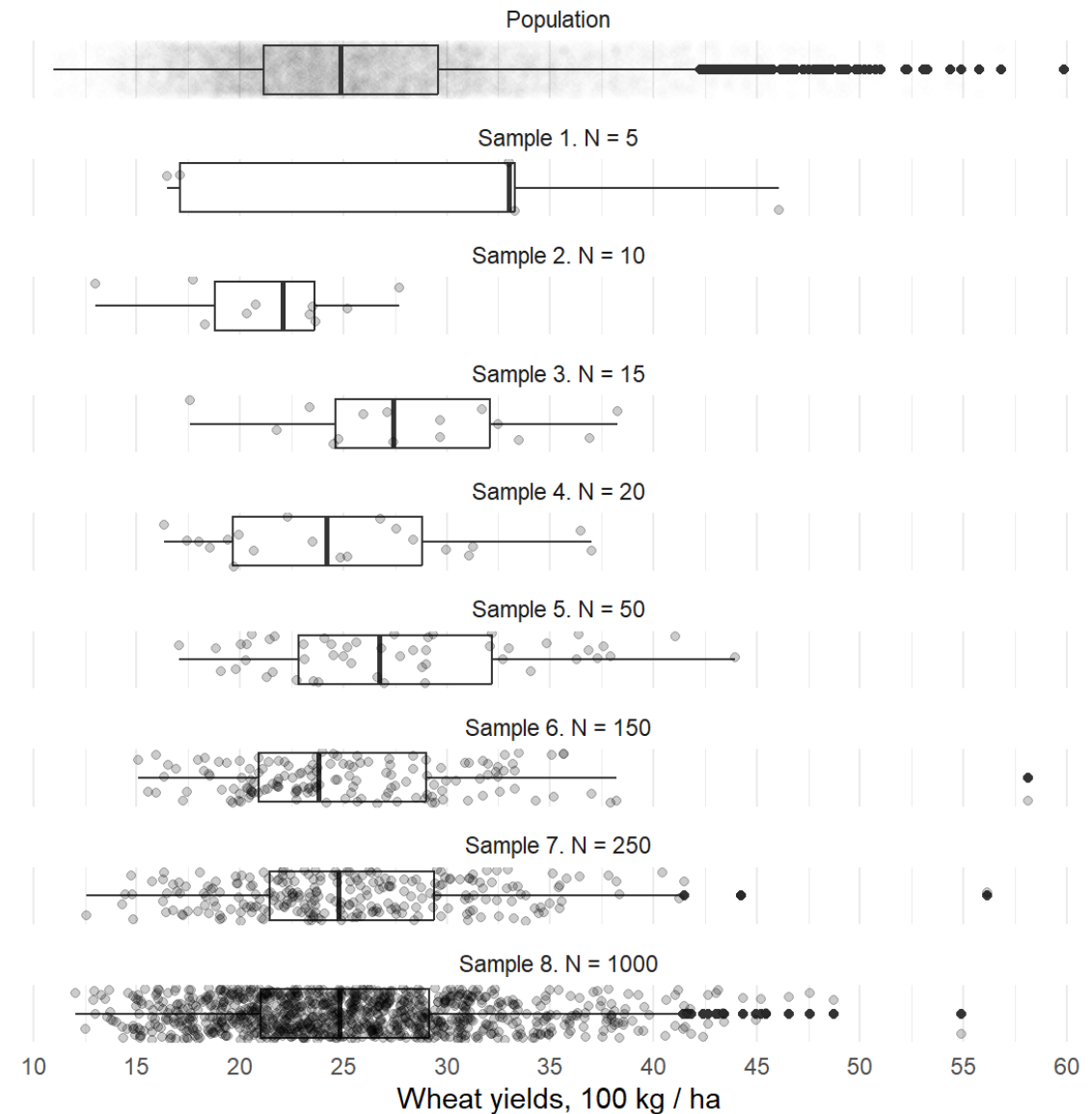
LLN makes random samples a powerful tool for understanding the population.

# Law of Large Numbers: action

Let us gradually increase the sample size!  
What will be difference between samples?

| Type2              | Yields |
|--------------------|--------|
| Sample 1. N = 5    | 29.19  |
| Sample 2. N = 10   | 21.36  |
| Sample 3. N = 15   | 28.31  |
| Sample 4. N = 20   | 24.72  |
| Sample 5. N = 50   | 27.54  |
| Sample 6. N = 150  | 25.22  |
| Sample 7. N = 250  | 25.44  |
| Sample 8. N = 1000 | 25.48  |

With increasing size, the sample becomes more alike the population!



# Law of Large Numbers: limitation

Sampling is costly: Larger sample size = larger expenses.

Each sample is still a random draw!

- Can we be sure that our sample estimates the right average?
- If a population has minorities, how accurately can a random sample represent them?

## Solutions:

- Apply appropriate sampling techniques!
- Embrace uncertainty and use inferential statistics.

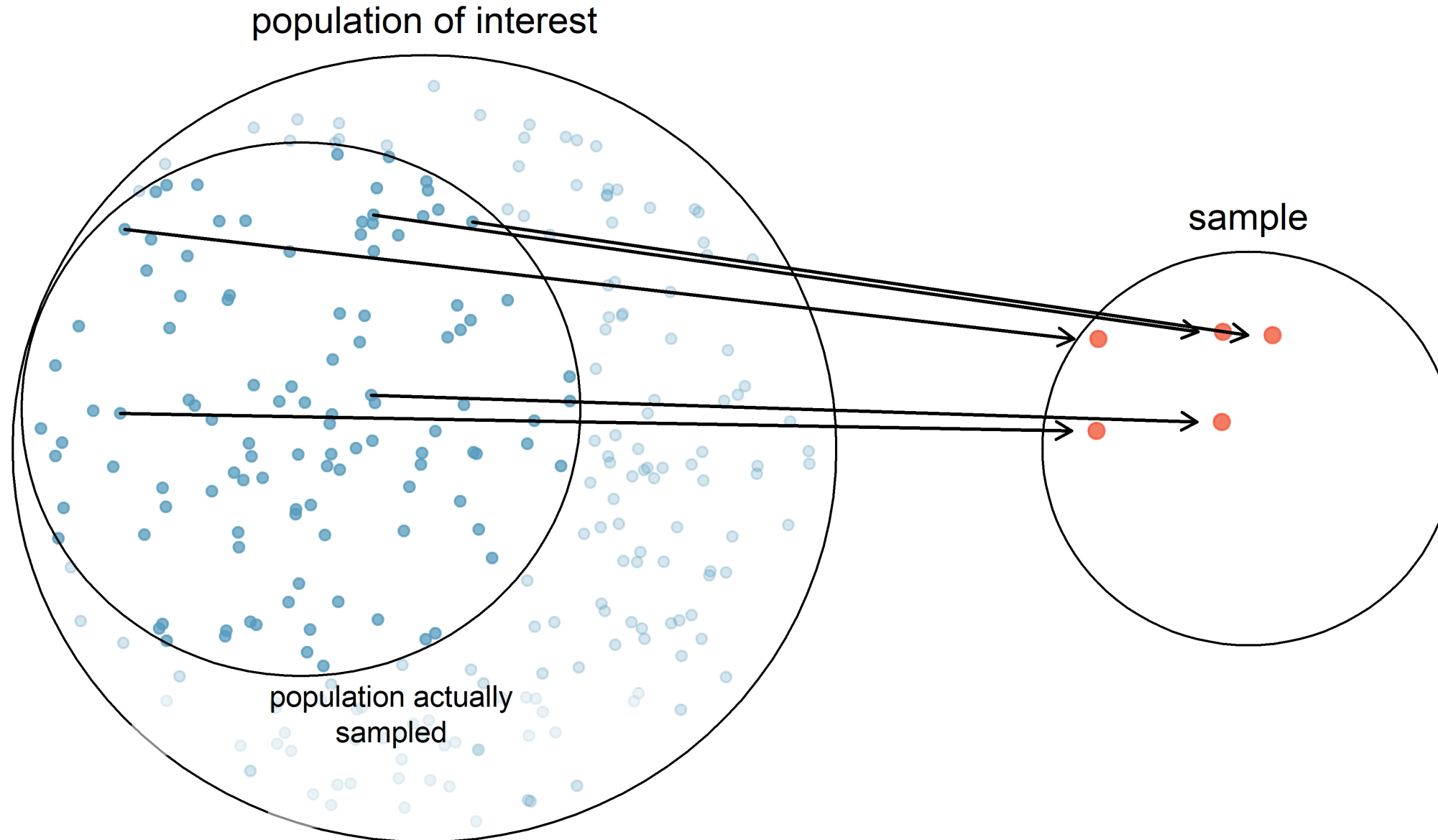
# Simple sample limitations (1)

Desired sample



# Simple sample limitations (2)

## Actual sample



# Simple sample limitations (3)

Key problems:

- The population may contain underlining structures: **strata / clusters**.
- Non-response may occur;
- Convenience-sample may take place:
  - where individuals who are easily accessible are more likely to be included in the sample.
- Any such limitation causes our sample to be not **representative** or to contain **measurement errors**.

# Solutions:

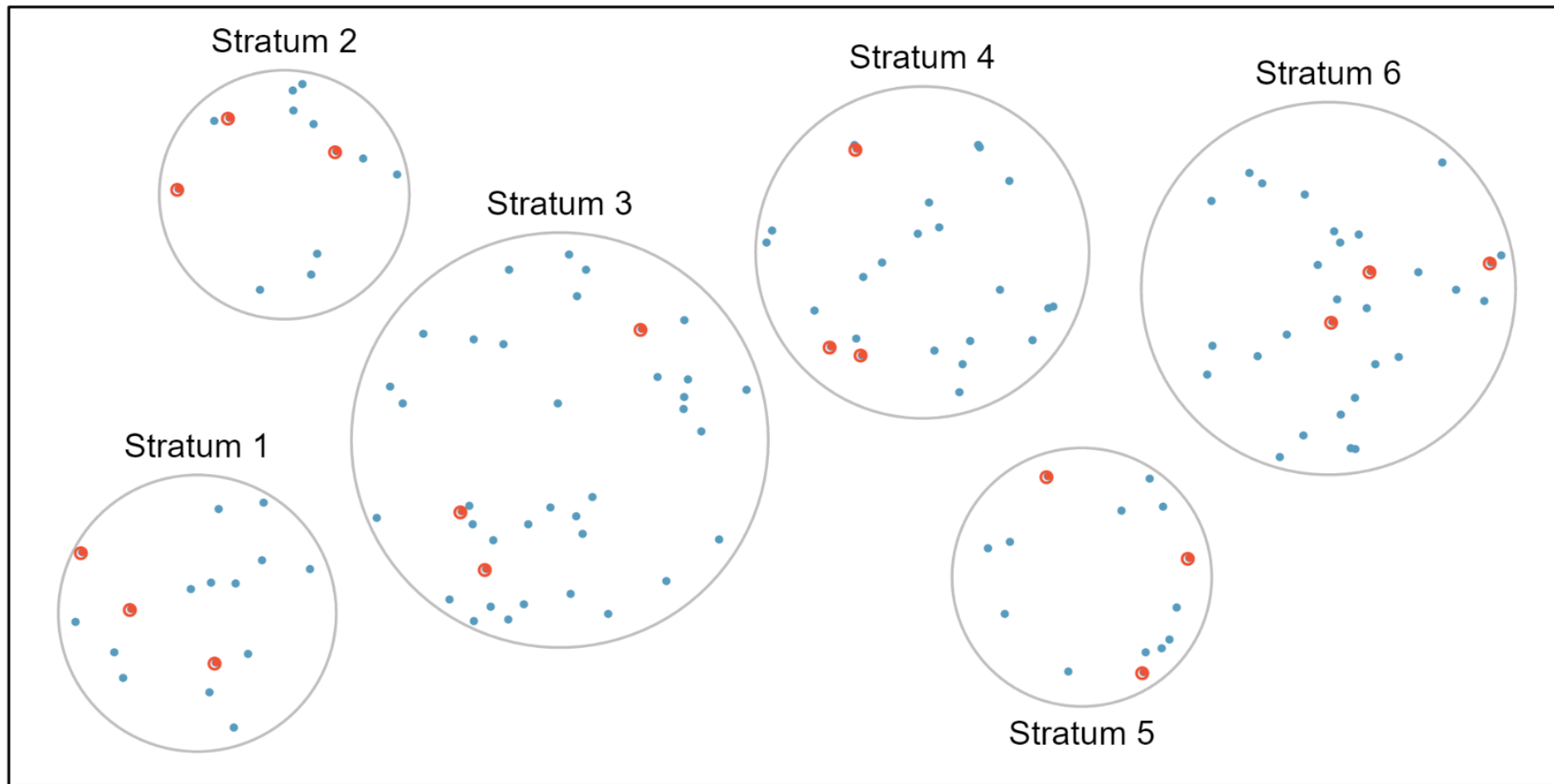
## Advanced sampling techniques

Those exist to balance between data quality and data collection costs.

- Simple;
- Stratified;
- Cluster;
- Multistage sampling;
- Population-weighted sampling and statistics;

To read more, see for example the book: Sampling by ([Thompson 2012](#))

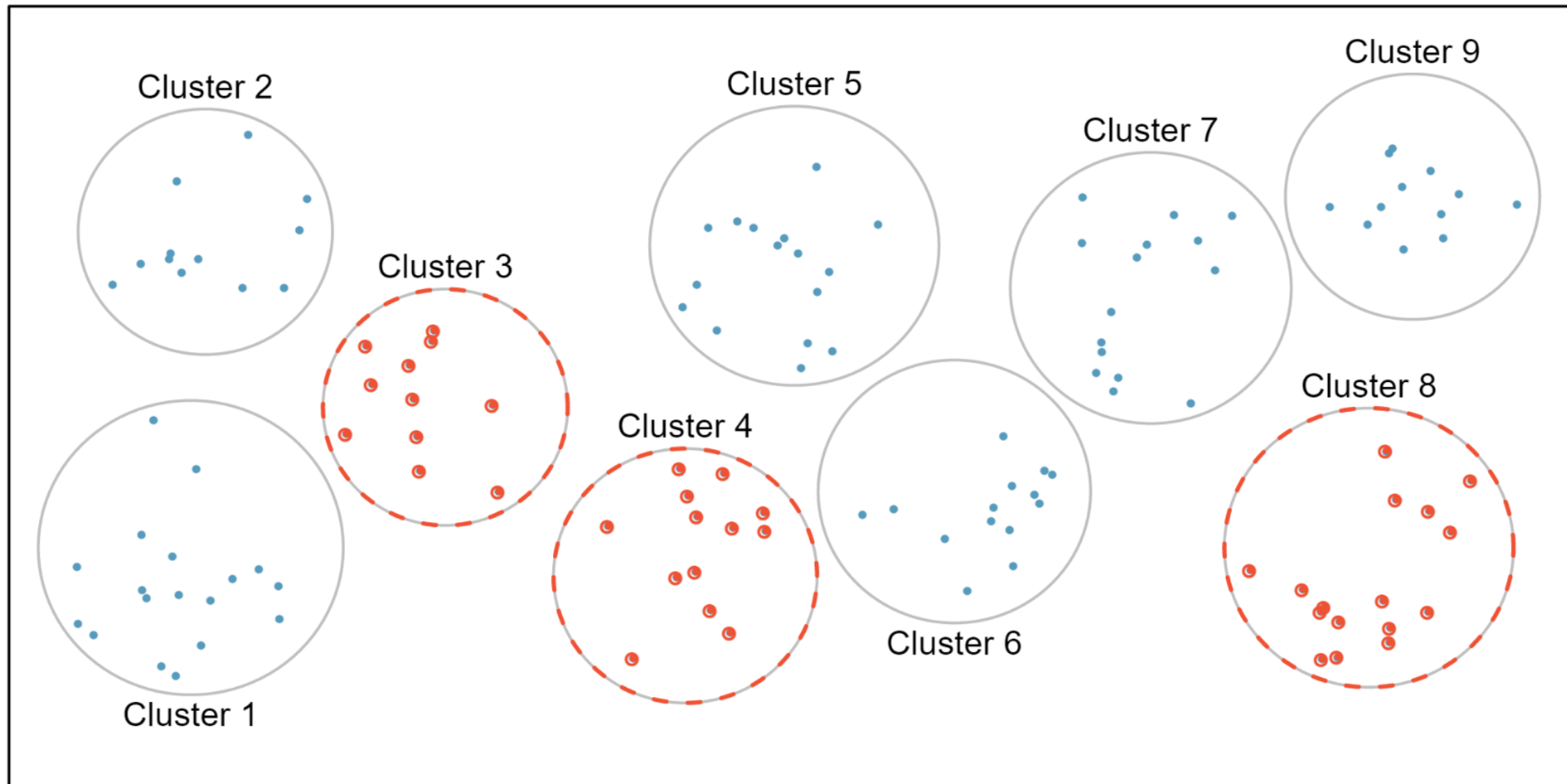
# Stratified sampling



The population is divided into groups **strata**.

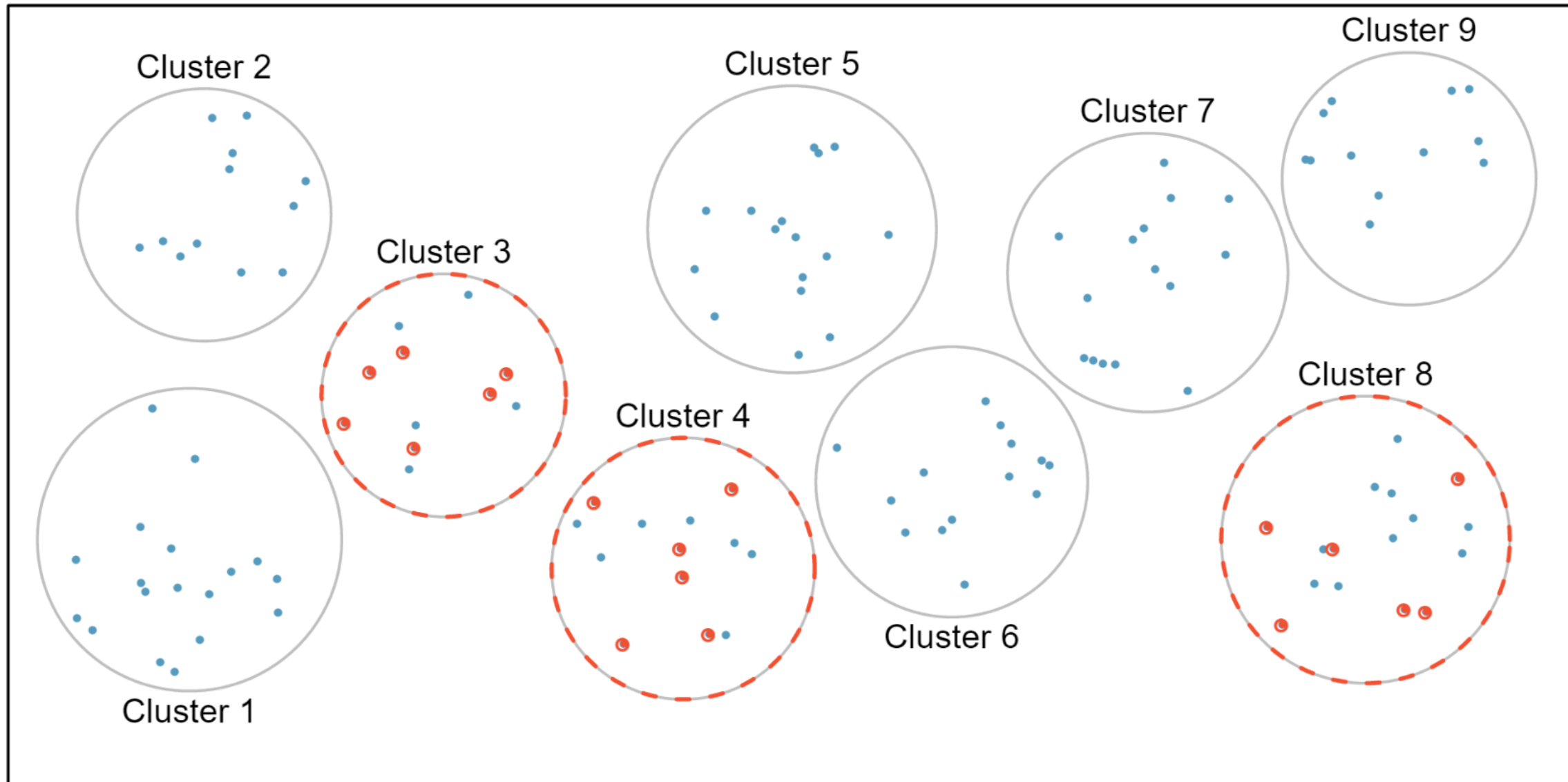
- strata combines **similar cases** especially with respect to response variables;
- within-strata: simple random sampling

# Cluster sampling



- Population is broken into clusters (by location usually).
- We randomly select several clusters and sample them completely.

# Multistage sampling



- Cluster + satisfied sampling
- Cluster + simple sampling

# Population and samples by research question

1. What is the average poverty headcount rate in the world/a country?

- Population: people or households;
- Sample examples: household surveys.
- Data source: The Living Standards Measurement Study (LSMS)  
<https://www.worldbank.org/en/programs/lsms> . . .

**At home:** discuss the same for other research questions mentioned earlier.

# Population and sampling: summary

- Research objectives are to conclude about the population.
- But the population is **never available**, instead we have **RANDOM samples**.
- Random samples are all different: result with different estimates.
  - Thus, descriptive statistics based on each sample is not identical to one based on the population.

# How to conclude about population using random samples and statics?

- Any ideas?
- There are:
  - Descriptive statistics
  - Inferential statistics

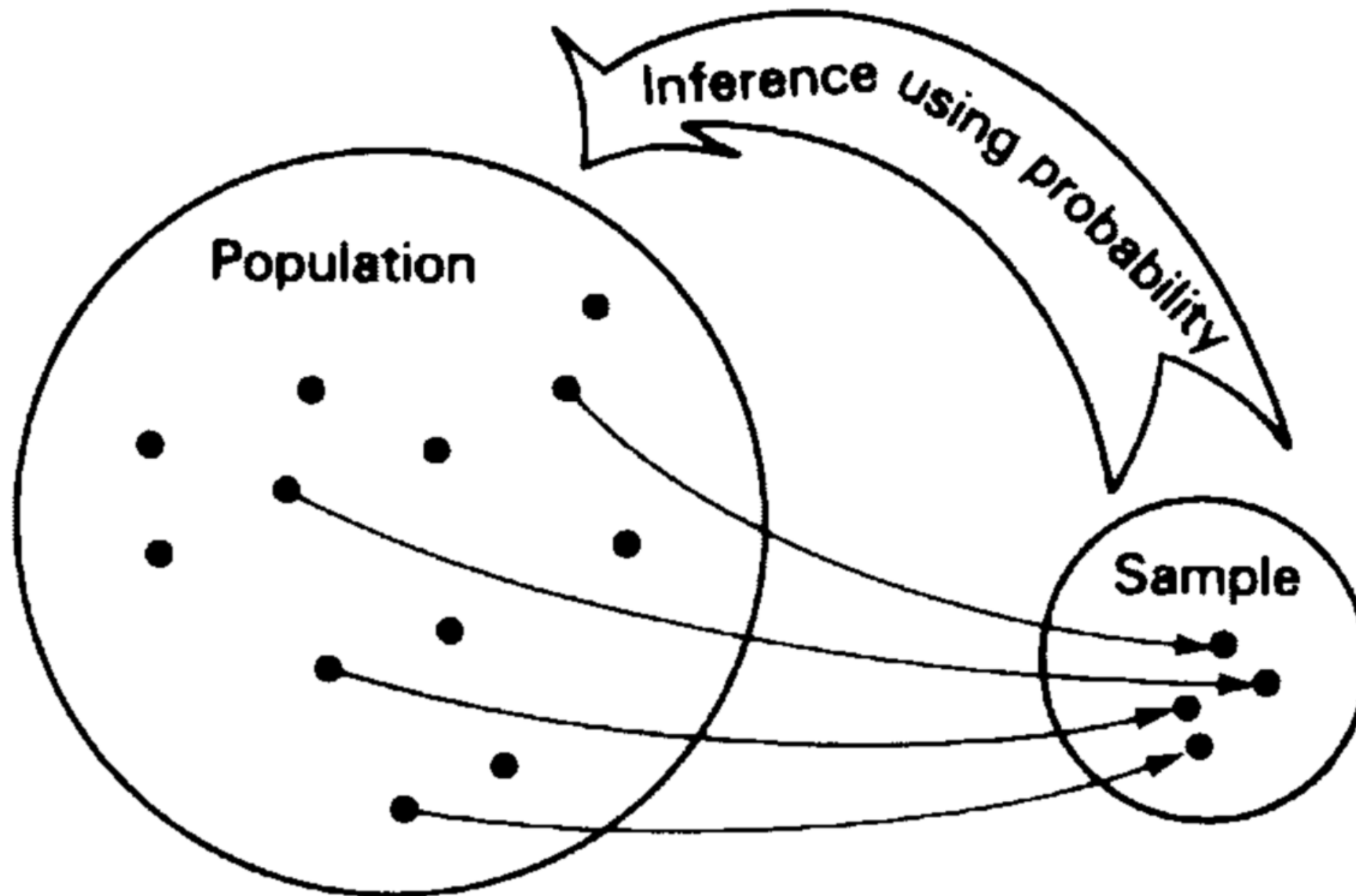
# Descriptive statistics

- AKA Data Analysis:
  - uses statistical methods to described observed data (data that comes from samples)
- But because each sample has a random component and is different from the population,
  - **descriptive statistics** does not tell us much about the **population**...

# Inferential statistics

- Answers questions **about the population** using:
  - the **sample**, and
  - **probability theory**
- The science of using probability to make decisions ([Dowdy, Weardon, and Chilko 2004](#))
- Does so by:
  - Establishing **statistical hypothesis**;
  - Using statistical methods and probability theory to **reject or accept** such hypothesis;

# The process of statistical inference



Statistical inference is based on **statistical hypothesis testing (HT)**.

# Statistical hypothesis: basics (1)

As we rely on a samples,

- we call any computed measure (such as mean, median, variance):
  - **the estimate of** {mean, median, variance}.
  - we denote estimates with a “hat” “ $\hat{\phantom{x}}$ ”
  - for example:  $\{\hat{\text{mean}}, \hat{\text{median}}, \hat{\text{variance}}\}$
  - Based on any sample we can **estimate** sample  $\hat{\text{mean}}, \hat{\text{median}},$   
 $\hat{\text{variance}}$

# Statistical hypothesis: basics (2)

To infer that {mean, median, variance} in the population is similar to **the estimate of**  $\{\widehat{\text{mean}}, \widehat{\text{median}}, \widehat{\text{variance}}\}$ , we:

## 2. formulate two statistical hypothesis:

- **H0** (the null hypothesis):
  - “The estimate of {... ..} is equal to ZERO in the population”.
- **H1** (the alternative hypothesis):
  - “The estimate of {... ..} is NOT equal to zero in the population”.

# Statistical hypothesis: basics (3)

3. use statistical tests and the probability theory to reject **H0** (or fail to reject H0) and conclude:

- We reject **H0** concluding **H1**:
  - with the probability  $P$  (p-value) that we've made a mistake by rejecting H0 (Type I error).
- We accept **H0**
  - because available data is insufficient to reject it;
  - without implying that H0 is the right hypothesis (Type II error);

# Goals of the Part 2: Quantitative Research Methods

- Grasp the difference between description and inference.
- Learn how to:
  - apply basic methods of descriptive statistics, and
  - perform statistical inference.
  - concludes based on their results.
  - communicate consistent stories

# At home: Example of statistical inference in a research

Scan through the example.

1. Research question
2. Research design
3. Data collection
4. Data analysis

# Research question

**Do extension services in agriculture improve rice yields in “a province” of Vietnam?**

- **The population: all rice producers in “a province” of Vietnam.**
- **The variables: rice yields in tons/ha and amount and quality of extension services received.**

# Research Design (simplified)

## Randomized Control Trial (RCT) experiment:

- Divide population into **clusters** and **strata**.
- **Clusters** are similar in structure but are located in different places: **villages**
  - Treatment group: 4 clusters with extension services.
  - Control group: 4 clusters without such.
- Same **strata** are present in each cluster: **small, medium, and large farmers**.
  - **Stratified Simple Random Sampling of 200 farmers in each cluster proportionally representing all strata.**
- Final sample size 1600 observations.

# Data Collection: Two waves

## Wave 1: pre-intervention

- Exist to establish the baseline;
- Establish the **check and balance tables**
  - To ensure that **on average, treatment and control groups are similar.**

## Wave 2: post-intervention

- Exist to check the hypotheses.
- Collects data on the variables of interest.

# Data analysis

1 year after the treatment, we repeat data collection and measure rice yields:

## Treatment group

Response rate: 1483

Rice yields: 6.0 tone per ha

## Control group

Response rate: 1542

Rice yields: 5.8 tone per ha

**Can we conclude extension services have a positive effect on the rice yields in Vietnam?**

Let us see what rice yields looks like in the world:

<https://ourworldindata.org/grapher/rice-yields>

To conclude about the effect of intervention, we need to

- “embrace” the uncertainty!
- perform statistical inference,
- consider **Variance** not only point-estimates.

# Takeaways checklist

# Homework



Study all prerequisites on your own: see Ilias.

- Familiarize with the World Bank DIME initiative  
<https://www.worldbank.org/en/research/dime>
  - And Dime knowledge base:  
<https://www.worldbank.org/en/research/dime>
  - Find one project on their website and try to understand what type of data collection was used there and how it was implemented.

# Key concepts covered in the lecture

- The population and a sample;
- Simple Random Sample: (non-) representative sample, convenience sample, measurement error.
- Other sampling technique: stratified, clustered and multistage sampling.
- Descriptive statistics.
- Inferential statistics.
- Exemplary research design.

# Example of the exam questions

- Why do we need samples and random sampling?
- Why random sampling is such a powerful tool?
- When does random sampling fail?
- What can descriptive statistics tell us about the population?
- Why do we need inferential statistics?
- When does simple random sampling fail?
- What key random sampling techniques exist and why do we need them?

# References

- Diez, David, Mine Cetinkaya-Rundel, and Christopher D Barr. 2022. *OpenIntro Statistics: Fourth Edition*. OpenIntro. <https://www.openintro.org/book/os/>.
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